

Circle One:

i-RAT*t*-RAT**CS 261 (Computer Systems I) – Binary Information RAT 2.1**

In this RAT, you will work in teams of 2-3 people each to explore and reinforce the material you've read in the textbook.

Name(s): _____

Part 1 – Base conversions

Fill in all missing values:

Binary (base 2)	Decimal (base 10)	Hex (base 16)
1	1 (2^0)	0x1
10	2 (2^1)	0x2
11	3 ($2^1 + 2^0$)	_____
_____	4	_____
1010	_____	_____
_____	23	_____
_____	_____	0x2C

Part 2 – Byte orderings

Fill in the following table with the little- and big-endian layout of the hex number 0x12345678 beginning at memory location 0xBF04. (You can leave off the “0x” in your answer.)

	Memory addresses			
	0xBF04	0xBF05	0xBF06	0xBF07
Big endian:	_____	_____	_____	_____
Little endian:	_____	_____	_____	_____

Circle the correct two-byte little-endian hex encoding of the decimal value 17:

11 00**00 11**

Circle the correct two-byte little-endian hex encoding of the decimal value 512:

02 00**00 02**

Part 3 – Bitwise operations

Perform the following bitwise operations and specify the answer in binary:

$$\begin{array}{r} 1110 \ 0110 \\ \& \ 1001 \ 0011 \\ \hline \end{array}$$

$$\begin{array}{r} 1110 \ 0110 \\ | \ 1001 \ 0011 \\ \hline \end{array}$$

$$\begin{array}{r} 1110 \ 0110 \\ \wedge \ 1001 \ 0011 \\ \hline \end{array}$$

Perform the following bitwise operations and specify the answer in 64-bit hex:

$$0x12345678 \ \& \ 0x00000000 = \underline{\hspace{2cm}}$$

$$0x12345678 \ \& \ 0xFFFFFFFF = \underline{\hspace{2cm}}$$

$$0x12345678 \ \& \ 0xFFFF0000 = \underline{\hspace{2cm}}$$

$$0x12345678 \ \& \ 0xF0F0F0F0 = \underline{\hspace{2cm}}$$

$$0x12345678 \ | \ 0x00000000 = \underline{\hspace{2cm}}$$

$$0x12345678 \ | \ 0xFFFFFFFF = \underline{\hspace{2cm}}$$

$$0x12345678 \ | \ 0xFFFF0000 = \underline{\hspace{2cm}}$$

$$0x12345678 \ | \ 0xF0F0F0F0 = \underline{\hspace{2cm}}$$

Bonus: Prove that the follow routine swaps "x" and "y" by symbolically expanding the results of each operation and using the properties of the XOR operator to simplify.

```
void swap (int *x, int *y)
{
    *y = *x ^ *y;
    *x = *x ^ *y;
    *y = *x ^ *y;
}
```

x

y

x

x ^ y
